LLM-Based Assertion Extraction: Prompt Design, Error Taxonomy, and Validation I

This supplementary section documents the implementation specifics of the LLM-based assertion extraction pipeline.

LLM-Based Assertion Extraction: Prompt Design, Error Taxonomy, and Validation II

Relationship to Prior Approaches

The closest prior effort is the systematic literature analysis of Knight, Cordes, and Friedman [knight2022fep], which used human annotators to manually code structural, visual, and mathematical features of FEP and Active Inference publications. Their work operated at the scale of hundreds of annotated papers and employed terms from the Active Inference Institute's Active Inference Ontology for automated text analysis. Our pipeline replaces the manual coding step with LLM-based assertion extraction, enabling scalable processing of the full corpus ($N = 817$ papers) at the cost of exchanging human-verified precision for machine-generated assessments that require post-hoc validation. This trade-off is characteristic of the broader LLM-based scientific extraction landscape: recent benchmarking confirms that even state-of-the-art modular extraction architectures fall short of production-level precision—particularly